

Advay Vyas

advay.vyas@utexas.edu | linkedin.com/in/advay-vyas | Austin, TX | (214)-603-8041

Education

The University of Texas at Austin, Austin, TX

August 2024 – May 2027

Bachelors of Science in Statistics and Data Science & Mathematics, GPA 3.95

Relevant Coursework: Real Analysis I & II, Probability, Stochastic Processes, Mathematical Statistics, Linear Algebra, Differential Equations, Software Design, Databases, Multivariable Calculus

Research Experience

Meyers Lab, The University of Texas at Austin, Austin, TX

March 2025 – Present

Worked primarily under Dr. Dongah Kim for my two projects.

Forecast Evaluation Project

- Evaluated local vs. aggregate infectious disease forecasts using probabilistic scoring methods, including the weighted interval score (WIS) and prediction interval coverage.
- Implemented forecast evaluation pipelines in R using the `scoringutils` package to compute and summarize WIS, including decomposition into overprediction, underprediction, and dispersion components.
- Developed comparative analyses using relative WIS to assess performance differences between local forecasts and per-capita or directly scaled aggregate baselines.
- Investigated associations between forecast accuracy and locality characteristics such as population size, density, and urbanization across multiple geographic strata and forecast horizons.
- Contributed to a forecasting hub study designed to evaluate sub-state infectious disease forecasting methods in the U.S. respiratory virus season.

Similarity Metrics Project

- Investigated whether local-level infectious disease forecasting provides meaningful improvements over state-level trends by quantifying divergence in epidemic curves for influenza, COVID-19, and RSV.
- Applied curve-based similarity metrics, including Discrete Fréchet distance and Dynamic Time Warping (DTW), to distinguish structural versus temporal differences between forecast trajectories.
- Extended trajectory comparison methods by implementing Soft-DTW and related similarity measures via `softdtw` and `similaritymeasures` to support more probabilistic comparisons.
- Analyzed early epidemic growth features (e.g., early slope, doubling time, second derivative, and week-to-week growth rates) to assess their association with peak timing differences between local and state-level epidemics.

Emory Computational Mathematics for Data Science REU, Atlanta, GA

June 2025 – Present

- Worked on the Hybrid Regularization for Random Feature Models project under Dr. Drum.
- Investigated hybrid regularization methods for solving ill-posed inverse problems.
- Implemented Hybrid LSLU with Random Feature Models in MATLAB with `IRTools`.
- Evaluated methods on MNIST, PneumoniaMNIST, and OrganMNIST3D datasets.
- Achieved performance comparable to LSLU while significantly reducing computational cost and runtime.
- Co-authored a manuscript and submitted to SIURO (SIAM Undergraduate Research Online).

Department of Mathematics, University of North Texas, Denton, TX

March 2023 – May 2024

- Conducted traffic flow research under Dr. Liu in the UNT Mathematics Department.
- Generated realistic traffic networks using `NetworkX` and modeled system dynamics using optimization techniques.
- Approximated gradient minima using Lagrange multiplier methods and solved with `NumPy`.
- Developed algorithms that reduced traffic irregularity by 31.3%.

Mixer Lab, University of North Texas, Denton, TX

October 2022 – May 2024

- Conducted neuroscience research under Dr. McMahan in the UNT Learning Technologies Department.
- Developed an educational game using Unity for experimental learning studies.
- Designed recursive experimental scenarios and supported EEG-based testing.
- Worked on tools aimed at helping educators tailor instruction to individual learners.

Publications

- McMahan, Timothy, Chotepong Victor Tangton, **Advay Vyas**, Chloe Huang, and Aron Yewoor (2024). “Bridging the Gap Between Perceived and Actual Personalities in Real and Virtual Worlds Through EEG Analysis”. In: *Annual Review of Cybertherapy And Telemedicine 2024*, pp. 206–212. ISSN: 1554-8716.
- Vyas, Advay**, Ethan Rebello, and Jianguo Liu (2023). “A Lagrangian Approach to Loss Function Optimization on Traffic Network Regularity”. In: *2023 IEEE MIT Undergraduate Research Technology Conference (URTC)*, pp. 1–5. DOI: 10.1109/URTC60662.2023.10534993.

Presentations

- “Martingales and Stopping Times,” Math Directed Reading Program Symposium, Spring 2026
- “Quantitative Metrics for Decomposing Timing and Magnitude Differences in Influenza Forecasts,” University of Texas at Austin Technology & Science Undergraduate Research Forum, Spring 2026
- “Inner Product Free Hybrid Regularization for Random Feature Models,” American Mathematical Society Joint Mathematics Meetings, Spring 2026
- “Hybrid Regularization for Random Feature Models,” Emory REU Poster Session, Summer 2025
- “Nowhere Differentiable, Everywhere Useful,” Math Directed Reading Program Symposium, Fall 2025
- “Topology and Data in Biology,” Math Directed Reading Program Symposium, Fall 2024
- “Infinity and Decisions,” Texas Academy of Mathematics and Science Rosecutting Ceremony, Spring 2024
- “A Dynamic Video Game to Assess Personality Complexes with EEGs,” University of North Texas Scholars Day, Spring 2024
- “A Lagrangian Approach to Loss Function Optimization on Traffic Network Regularity,” MIT IEEE Undergraduate Research Technology Conference, Fall 2023

Awards & Achievements

- College of Natural Sciences Advanced Summer Research Fellowship, University of Texas at Austin, 2026
- Best Use of Statistical Analysis, American Statistical Association DataFest at SMU, 2026
- Score of 3, William Lowell Putnam Mathematical Competition, 2025
- College of Natural Sciences Honors Summer Scholarship, University of Texas at Austin, 2025
- National Merit Finalist Scholarship, National Merit Scholarship Foundation, 2024
- Polymathic Scholar, University of Texas at Austin, 2024
- Eagle Scout, Boy Scouts of America, 2023
- Undergraduate Research Fellow Scholarship, University of North Texas, 2023